Received: January 10, 2023 | Accepted: February 08, 2023 | Published: February 15, 2023

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ORIGINAL RESEARCH

*Peercite* Journal of Infectious and Rare Disease

# PARASITIC INFECTIONS IN PATIENTS WITH TYPE 2 DIABETES MELLITUS IN BENGHAZI, LIBYA

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**DOI :** https://doi.org/10.61641/pjird.2023.1.1.4001-4004

## Abstract :

Diabetes is becoming a first-rate continual disorder burden all around the global. Little is understood about the prevalence of intestinal parasitic infections in them. The goal of this have a look at turned into to decide this occurrence in kind 2 diabetes mellitus in Benghazi, Libya. A cross sectional case control study recruited hundred kinds 2 diabetics from Benghazi diabetic center and one hundred non-diabetic people which served as manage institution become taken from Benghazi medical middle and Elhiaa sanatorium. A while ranged between 18 to 55 years with and without diarrhea between January and December 2016. Socio demographic characteristics had been taken for the duration of the interview. Fresh stool specimens had been gathered and processed, and data changed into analyzed. The prevalence of intestinal parasites turned into better amongst diabetic sufferers (40.0%) than among on diabetic sufferers (16.0%). Mainly, the following intestinal parasites were particularly typical in diabetic topics Entamoeba coli (10.5%), Cryptosporidium parvum (8.5%), at the same time as in control group Entamoeba histolytica (10%), Ascaris Lumbricoides (3%) were the maximum remoted parasite. There was a full-size dating among incidence of parasite and diarrhea ($p = 0.001$), schooling ($p = 0.000$), single and more than one infections ($p = 0.001$). Relation was not vast among parasitic infections with sex and age in all sample ($p = 0.203$, $p = 0.56$) respectively. The present observe emphasizes the need of increasing awareness among clinicians regarding the incidence of intestinal parasites at higher occurrence in kind 2 diabetic patients.

**Keywords :** Intestinal parasites, Diabetes mellitus

## Introduction :

Intestinal parasites are two foremost kinds specifically helminthes and protozoa which might be crucial causes of infections in immunocompromised people. In healthy man or woman commonly they cause less aggressive sickness. It's a public health hassle specifically in tropical nations where bad sanitary situations and contamination of meals and water assets. The developing countries are more vulnerable to intestinal and further-intestinal parasitic illnesses. Even in nicely evolved vicinity a number of these parasites can be in purpose vast ailment in the immunocompromised patients and young children. About 340 parasites infect an anticipated 3.5 billion humans and motive clinical signs and symptoms and symptoms in approximately 450 million.

Diabetes mellitus is this type of metabolic sicknesses that could have an effect on immune gadget.

Diabetic patients were been pronounced to be the immunocompromised which predispose them to many of infection which include parasitic infection.

The superiority of the diabetic patients infected with intestinal parasite in our use is nearly undetected and the efficacy of the treatment or preventive methods are difficult to understand that's why we conduct this look at to determine the burden of parasitic infection in each diabetic comparing to non-diabetic Benghazi populace.

## Materials and methods:

## Design and sufferers:

A go sectional case control takes a look at recruited two hundred kinds 2 diabetics (134 females and 66 adult males) from Benghazi diabetic center and 100 non-diabetic individuals (73 ladies and 27 men) which served as manage institution become taken from Benghazi clinical middle and Elhiaa clinic a long time ranged between 18 to 55 years with and without diarrhea between January and December 2016. Socio demographic characteristics have been taken all through interview. Fresh stool specimens had been collected and processed. The glycated hemoglobin (HbAlc) turned into measured to detect DM in manage organization.

## Specimen collection and processing:

All people in this look at were interviewed by using the use of a questionnaire. Stool samples were collected in a dry, smooth, leak proof plastic box have been transported to the studies laboratory of Institute Medical Professions-Benghazi without delay. Half of each pattern was stored in 10% formalin to prevent bacterial action. These samples were examined macroscopically through naked eye for appearance, coloration, consistency and presence of blood mucous and person or larvae helminthes microscopically following direct and formalin-ether attention method. Direct microscopy of the smears in a saline (0.85% NaCl solution) to verify the presence of motile intestinal parasites and trophozoites beneath light microscope (40x magnifications) and Lugol's iodine become carried out for the detection of ova, larvae, trophozoites and cysts of intestinal parasites.

Similarly, an attention technique was hired that involved mixing the stool samples with formalin, treating with ether, centrifuging afterwards to growing the threat of detecting protozoan parasite oocysts.

The layers of ether, formalin, and particles have been discarded and the residues were used to research for the presence of intestinal parasites, also we used changed Ziehl-Neelsen staining method for detection of oocysts of opportunistic coccidian intestinal parasite Cryptosporidium spp, Isospora belli and Cyclospora cayetanensis.

The information changed into analyzed using SPSS version 17 statistical bundle. A contrast of the frequency of parasites between cases and controls became carried out by chi-square check.

## Results:

Prevalence of intestinal parasites turned into forty.0% amongst diabetic better than non-diabetic 16.0% with high significance ($\chi 2$ = 37.576, df = 10, p=0.000). Parent 1 show this incidence in diabetic and non-diabetic player.

Specially, the following intestinal 10.5% of two hundred general diabetic/26.3% of infected diabetic patients, Cryptosporidium parvum 8.5% of 2 hundred general diabetic/21.3% of infected diabetic, even as on top of things organization Entamoeba histolytica (10% of one hundred general non-diabetic/60 3% of infected non diabetic), Ascaris lumbricoides 3% of 100 general non-diabetic/19% of inflamed non diabetic had been the maximum isolated parasite.

Desk display extraordinary sorts of parasite remoted from diabetic and non-diabetic manipulate and their percent calculated from total inflamed variety (Table 1).

| Type of parasites | Infection (%) | |
|---|---|---|
| | Patients with DM (n=80 ) | Control (n=16) |
| Blastocystis hominis | 1.3 %(1) | 6.3 %(1) |
| Entamoeba histolytica / E. dispar | 12.5% (10) | 63.0 %(10) |
| Giardia lamblia | 12.5%(10) | 13 %(2) |
| Entamoeba coli | 26.3% (21) | 0 %(0) |
| Cryptosporidium parvum | 21.3% (17) | 0 %(0) |
| Entamoeba hartmanni | 11.3% (9) | 0 %(0) |
| Isospora belli | 6.3% (5) | 0 %(0) |
| Dientamoeba fragilis | 3.8% (3) | 0 %(0) |

**Table 1**: Prevalence of different intestinal parasitic infection among infected DM and non DM control.

DM comparison between intestinal parasitic infections and duration of type 2 diabetes in patients. This study showed that the highly statistically significantly (p = 0.001) between the Duration of DM with the intestinal parasitic infections. ($\chi^2$= 40.212, df =1, p = 0.001) (Table 2).

| Total | Parasitic negative | Parasitic positive | Duration of DM (year) |
|---|---|---|---|
| Less than 5 year (n=105) | 30\105 (28.6%)<br>30\80 (37.5%) | 75 | 105 |
| ≥5 (n=95) | 50\95 (52.6%)<br>50\80 (62.5%) | 45 | 95 |
| Total | 80 | 120 | 200 |

**Table 2:** Duration of type 2 diabetes in patients and parasitic infections.

The relation between the diabetes status and the hazard parasitic contamination changed into strongly good sized ($\chi2$ = 17.64, df = 1, p = 0.000, odds ratio = 3.5, CI = 1.94-6.4). There has been a widespread relationship between occurrence of parasite and history of diarrhea ($\chi2$ = 26.5, df = 1, p = 0.000, Odds ratio 5.94, CI = 2.84-12.4) as well known and as separate for diabetic (p-value = 0.001) and non-diabetic (p-value = 0.000) (figure 1).

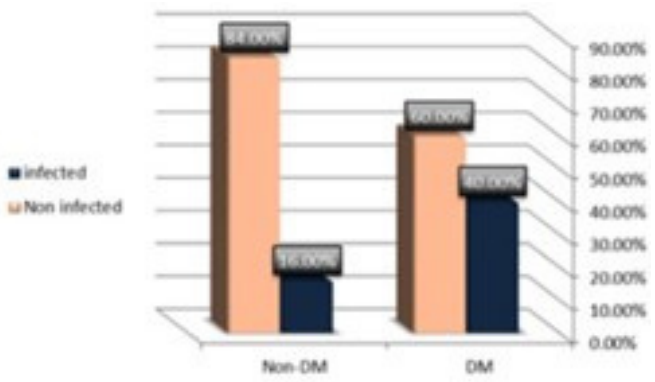


**Figure 1:** prevalence of parasitic infection in diabetic and non-diabetic participant.

## Discussion:

The prevalence of parasitic infection in our study turned into 40% in diabetics which considerably better than in non-diabetic 16%, this end result has been in concordance with some previous research, their take a look at in Iran reveal that the rate of parasitic infection in diabetics changed into 26.3% that is higher than manage 6.8%. Additionally, in settlement with which can be defined by way of the disorder in immune reaction in diabetic patients. In evaluation at Cameroon and at south east turkey, their studies display decrease incidence of parasitic infection amongst diabetic (10%, 47%) evaluating to non-diabetics (23.5%, 55.9%,) respectively.

They defined their end result via truth that the more variety of health practitioner visits incurred through diabetic sufferers than the non-diabetic sufferers in which diabetic patients seek advice from frequently and are treated for feasible intestinal parasitic infections. Of observe each final studies have been carried out at areas recognized to have excessive parasitic contamination fee, whilst Olusegun et al 2013conduct the have a look at place with decrease rate of parasitic infection. So it appears in areas have been endemic for parasite the impact of diabetes attenuated.

## Conclusion:

The relation among the diabetes fame and the hazard of parasitic infection turned into strongly considerable in us examine (OR = 3.5) i.e. the diabetes increases the danger of parasitic infection which assessment with which display lower risk with diabetes. A few previous studies display no distinction as regard to sex, age and this in concordance with our result for all pattern (p cost = 0.203) and in diabetics. Even as many studies display that girl and older a while has better incidence of parasitic contamination this just like our end result for non-diabetic (P fee = 0.04 for sex, p price = 0.001 for age). So it appears diabetic fame attenuate the impact of age and sex at the threat of acquiring parasitic contamination. Lower schooling level and high quality history of diarrhea was sturdy predictor for parasitic infection specifically mixed contamination and this in settlement with some of preceding studies.

## Conflict of interest:

Author declares there is no conflict of interest.

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**Citation:** Younis EZ et al. (2023) Parasitic infections in patients with type 2 diabetes mellitus in Benghazi, Libya. Peercite J Infect Rare Dis. 1(1):4001-4004

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